Who Bears the Load? Disparities in Childcare Time Across Family Structure, Gender, and Socioeconomic Factors in the U.S

Jasmine Chen '27 | Data Science Capstone Project

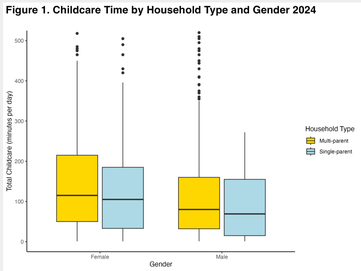
Background & Research Question

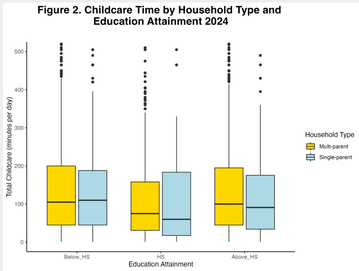
- Due to gender norms, stereotypes, and social expectations, the time parents spend on child care can vary based on family dynamics.
  - These variations can often imply underlying social values and biases in parental care.
- Child care is also a crucial factor to a child's growth and their physical and mental health. Based on previous literature (Kalenkoski et al. 2007), it is evident that family structure can have an impact on the time spent on child care.
- In this project, we attempt to use the American Time Use Survey to study the impact of family structure on time spent on child care in more recent days using newer data and compare whether similar trends are still observed, and whether there has been a change over time.

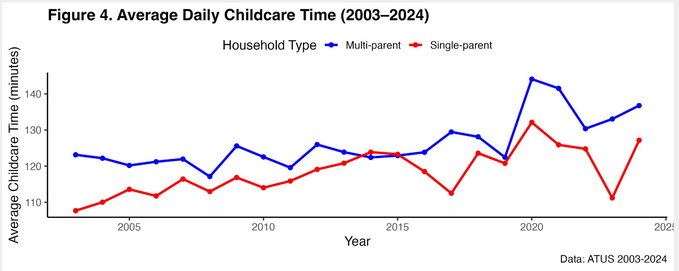
Research Question:

How does the time spent on unpaid childcare differ between single parents (households with only one adult and at least one child) and parents with other members in their household in 2024, and how does this relationship vary by employment, gender, education, and region? How has the trend changed over time since 2003?

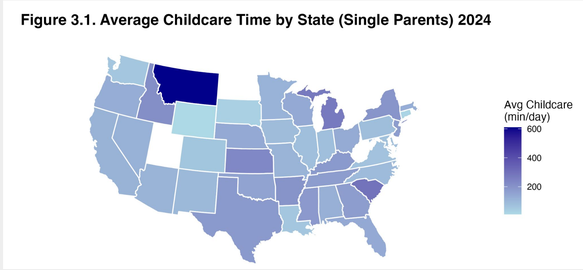
Visualizations

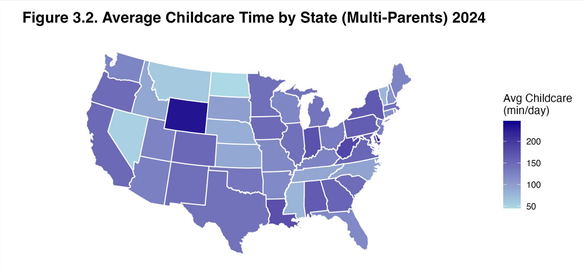






Female parents generally spend more time on childcare compared to male, and the childcare time for multi parents is slightly greater for both gender groups (Figure 1). It seems like educational attainment matters more for single parents, as there are more variations between groups (Figure 2).





Montana has a drastically higher average for childcare time for single parents in 2024 (Figure 3.1). The same pattern is not observed for multi parents (Wyoming instead) (Figure 3.2). From 2003 to 2024, the average childcare time for single parent families is lower than that for multi parent families for most years except for 2014 and 2015 (Figure 4). The trend between the two seems to be moving simultaneously.

Methodology

Data

This study uses data from the American Time Use Survey (ATUS) conducted by the Bureau of Labor Statistics, combining the single-year 2024 file (n = 7,669) and multi-year 2003–2024 file (n = 252,808) to examine childcare patterns and trends over time.

Data Cleaning

- Single parents were identified as respondents with at least one own child and no spouse or partner in the household.
- Total daily childcare and work time were computed by summing activity durations from the activity file by activity code.
- Demographic variables including education attainment and region were merged in from the CPS file, and missing values were imputed with zeros.

Modeling

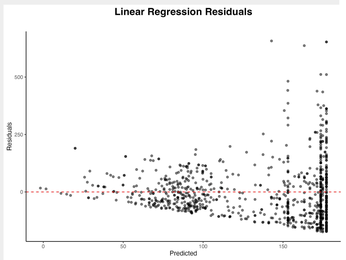
- A VIF test was conducted prior to model fitting to identify potential multicollinearity within predictors, and Gender and Region (6 specifically) were removed as a result in the MLR model.
- Two predictive models were estimated on the 2024 data using an 80/20 train-test split with k-fold cross-validation (k = 5).
- Model 1: Multiple linear regression (MLR) with total childcare time as the outcome and single parent status, work time, and education attainment as predictors.
  - Constructed residual and normal Q-Q plot to verify model assumptions
- Model 2: Random forest regression (500 trees) using the same predictors plus gender and region.
  - Constructed residual plot for model performance evaluation
- Model performance was compared using RMSEs and k-fold CV scores on the held-out test set.

Models

Multiple Linear Regression Model

$$\hat{Y}_{childcare} = \beta_0 + \beta_1 X_{single_parent} + \beta_2 X_{work_time} + \beta_3 X_{EducationAttain} + \epsilon$$

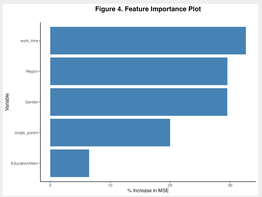
	Estimate	Std. Error	t value	Pr(>|t|)	Significance
(Intercept)	177.168	3.352	52.860	0.000	***
single_parentSingle-parent	-2.329	6.080	-0.383	0.702	
work_time	-0.172	0.008	-21.503	0.000	***
EducationAttainHS	-24.049	5.859	-4.105	0.000	***
EducationAttainAbove_HS	-3.544	4.222	-0.839	0.401	

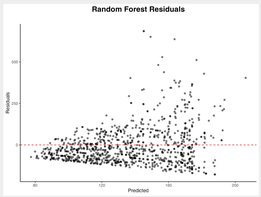


RMSE: 119.7718 CV score: 14796.37

Random Forest Model

	Importance	
	%IncMSE	IncNodePurity
single_parent	19.992961	143930.5
work_time	32.627400	4444079.3
Gender	29.510536	614281.0
EducationAttain	6.483384	289526.9
Region	29.526788	3786524.7





RMSE: 116.3911 CV score: 15420.25

Conclusion & Discussion

Conclusions

- On average, multi-parent household mothers tend to spend the longest amount of time on childcare, which can potentially be explained by being stay-at-home wives and gender role expectations. Single parents may have more financial burden that requires them to work more to support the family, resulting in less time with their child. However, it is important to note that our study result contradicts previous literature, potentially due to the variation in single-parent definition.
- Parents with high school degree seems to spend the least time in childcare for both groups, and there exists some regional differences in average childcare time, but these patterns are not the same for single- vs. multi-parent households. More Southern states seem to have greater average childcare time for single parents. Yet, there is not a definitive way to explain the regional variations.
- As expected, there is a negative correlation between childcare time and work time, suggesting that parents who spend more time in work spend less time with children.

Limitations

- The performances of both models are not ideal, with relatively high RMSEs. The equal variance assumption is likely not met from the pattern of the residual plots.
- Due to the lack of information, the way of defining single parents may not be fully accurate, and there are other factors that are not considered, including childcare support from other family members that don't live in the same household, intangible time on childcare, divorced/co-parenting parents, etc.

Future Work

- Future studies should focus on exploring the nuance between these other factors can our study cannot include to accurately classify single parents.
- Further studies need to be done to find the best models for predicting childcare time with more predictors.

Acknowledgement

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References

Kalenkoski, C. M., Ribar, D. C., & Stratton, L. S. (2007). The effect of family structure on parents' child care time in the United States and the United Kingdom. *Review of Economics of the Household, 5*(4), 353–384. https://doi.org/10.1007/s11150-007-9017-y